

# Abdullah U. Mustafa

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<https://aumustafa.github.io>

## Education

### (Medical) Physics

University of Windsor

Honours Bachelor of Science

Minors in Mathematics and in Chemistry

Windsor, ON

Sep. 2023 – Apr. 2027

## Research Experience

### Ultrashort Laser Pulse Measurements

May 2026 – Present

*Attosecond Condensed Matter Experiments Lab*

Main focus is frequency-resolved optical gating via Kerr instability amplification. Performed experiments, data acquisition, reconstruction and other analysis, wrote software, and wrote simulations. The long-term goal for this project is to eventually measure damage effects/plasma density in crystals wrought by KIA.

### Laser-Induced Breakdown Spectroscopy on Bacterial Cells

Oct. 2023 – Present

*Rehse Lab*

The main focus is to develop a quasi-instantaneous diagnostic test for bacterial diseases, such as bacterial meningitis, using laser-induced breakdown spectroscopy. Performed experiments, acquired data, applied chemometric data analysis and machine learning techniques for classification, fixed lasers, and prepared biological samples.

## Publications

- Nathan G. Drouillard, **Abdullah U. Mustafa**, TJ Hammond. “Kerr-based parametric amplification of polarization-gated cross correlation frequency-resolved optical gating.” *Optica Open*, preprint, 2026. <https://doi.org/10.1364/opticaopen.34003644>

## Recent Presentations

- *Establishing the limit of detection of laser-induced breakdown spectroscopy for Escherichia coli in artificial cerebrospinal fluid for the diagnosis of bacterial meningitis.* **Abdullah U. Mustafa**, Lauren Dmytrow, Rachel Chevalier, Jasmine Fric, Simona Brezeanu, Emma Pesce, Katherine Keller, Jack Holland, Will Conlon, Danielle Renard, Faiza Fric, Ayman Younes, Steven Rehse. Canadian Association of Physicists Congress 2026, “Quantitative Biomedical Imaging and Sensing I.” June 23, 2026. (Talk delivered by A. Mustafa.) [Link to slides.](#)
- *Femtojoule Pulse Reconstruction via Kerr Instability Amplification.* Nathan G. Drouillard, **Abdullah U. Mustafa**, T. J. Hammond. Canadian Association of Physicists Congress 2026, “Ultrafast Phenomena.” June 22, 2026. (Talk delivered by T. Hammond.) [Link to slides.](#)
- *Differentiation of bacterial species in biomedical specimens using laser-induced breakdown spectroscopy.* Lauren Dmytrow, **Abdullah U. Mustafa**, Rachel Chevalier, Matteo Pontoni, Simona Brezeanu, Jasmine Fric, Katherine Keller, Emma Pesce, Steven Rehse. Canadian Association of Physicists Congress 2026, “Biophysical Approaches to Structure, Function, and Environmental Response.” June 22, 2026. (Talk delivered by L. Dmytrow.)
- *LIBS for global health: detecting and diagnosing pathogenic bacteria in human clinical specimens of blood, urine, and cerebrospinal fluid.* Steven J. Rehse, Rachel Chevalier, **Abdullah U. Mustafa**, Lauren Dmytrow, Simona Brezeanu, Jasmine Fric, Emma Pesce. Pacificchem 2025. The International

Chemical Congress of Pacific Basin Societies 2025, Honolulu, Hawaii, December 15–20, 2025. (Invited talk delivered by S. Rehse.)

## Awards

- NSERC Undergraduate Student Research Award (2026)
- Dr. Lucjan Krause Undergraduate Scholarship in Physics (2025)
- Outstanding Scholars Award (2024–2026)
- President’s Entry Scholarship (2023)
- Dean’s List (2023, 2024, 2025)
- 1<sup>st</sup> Place Winner in the Canadian Space Agency Hackathon (2023)

## Skills

- Optics and Laser Experience
- Python for data acquisition, spectroscopic/chemometric data analysis, machine learning, writing simulations, and **writing packages**. Java, C, HTML/CSS, and  $\text{\LaTeX}$  (this document).
- Poster and oral talk design and presentation skills.
- 3D modelling and printing.

## Other Activities

**Co-president/Co-founder of Science-Focused Research/Journal Club** Jun. 2026 – Present  
*“Pulse of the Page”*

Established and then ran a reading club where participants read and presented science papers of their choosing, in order to practice reading, presenting, and critically analyzing scientific literature.

**Teaching Assistant** Sep. 2025 – Mar. 2026  
*Department of Physics*

Led (taught, instructed, graded) a pair of introductory physics labs on mechanics and electromagnetism.

**Physiotherapy Volunteer** Jun. 2025 – Apr. 2026  
*Hôtel Dieu Grace Healthcare*

Volunteered at the physiotherapy floor at a local hospital; assisted patients with their exercises.

**Leadership Positions** May 2025 – Apr. 2026  
*“Outstanding Scholars Student Council” and “Physics Club”*

Managed outreach and organized events on different student groups.

**Mentor** Sep. 2024 – Present  
*“Outstanding Scholars” Mentorship Program*

Mentored multiple first-year students through an organized mentorship program.

**Cancer Charity Chair** Apr. – Jun. 2023  
*“Relay For Life”*

Managed a charity raising money for the **CCS**; supervised different subcommittees and ran meetings.